License Plate Detection and Recognition on CCPD2019

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Plan

Question

Do these changes improve end-to-end plate recognition?

- Task: detect and read Chinese license plates
- Baseline: frozen YOLO11s detector + EasyOCR
- Upgrade 1: fine-tune the detector on CCPD2019
- Upgrade 2: replace EasyOCR with a task-specific CRNN

Baselines

Detection: frozen YOLO11s

- Pretrained plate detector, 640 px, top-score box only
- Weights `license-plate-finetune-v1s.pt`

Recognition: EasyOCR

- General-purpose OCR engine
- No Chinese plate prior

CCPD2019

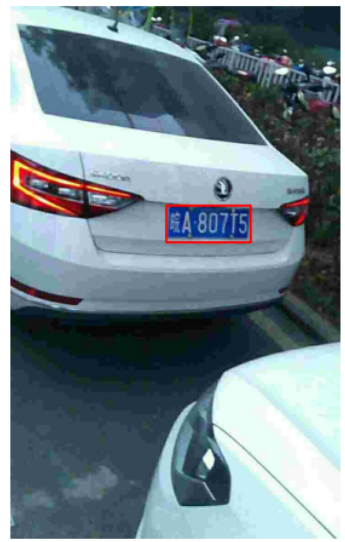
large, 1.8°



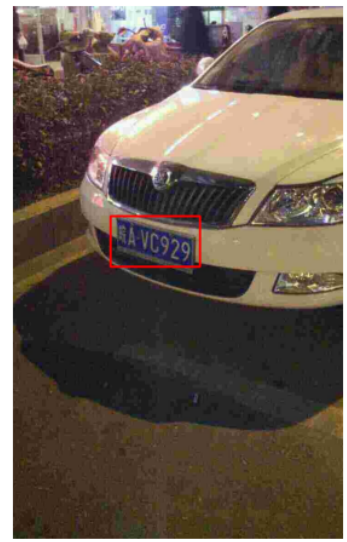
large, 10.3°



small, 9.1°



medium, 10.2°



Data Preparation

The dataset

- 352k images
- Box, tilt angles and plate text encoded in the file name

Our subset

- Stratified sampling by plate size and tilt angle
- 5,000 images
- Train / Test / Validation split: 3,500 / 700 / 800

Alphabet

Province character + latin letter + 5 alphanumeric characters; 7 characters per plate.
Training alphabet is 58 characters: 24 province characters + 10 digits + 24 latin letters (I and O excluded).

Stratification strategy

Goal

Preserve the source distribution of plate size and tilt angle: results reflect real conditions, not an easy subset.

Strata

- Plate height × tilt
- Quotas proportional to stratum size
- Split 3,500 / 700 / 800 assigned within strata, before scale variants

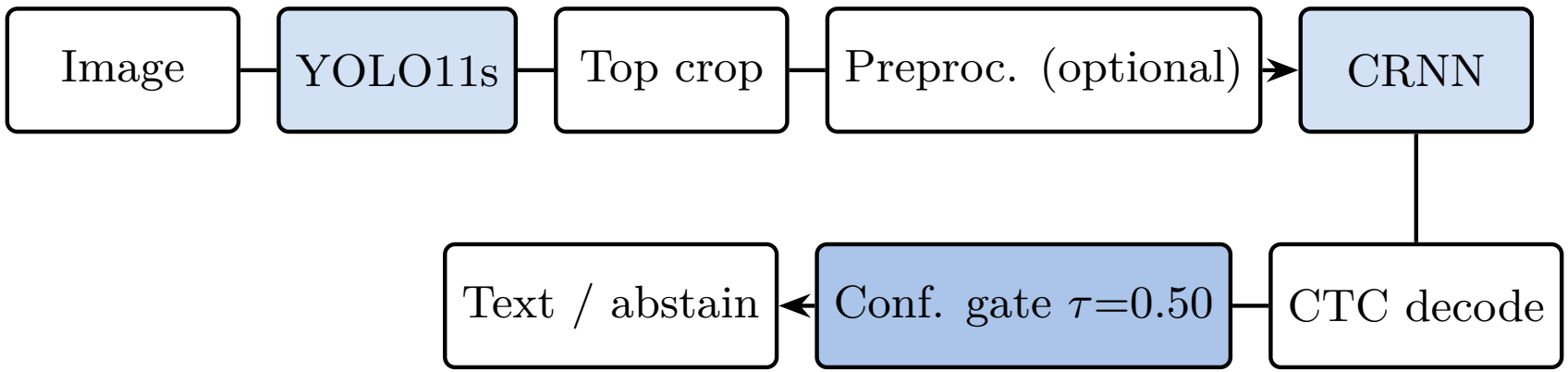
Scale variants

- 1.0 / 0.75 / 0.5 per image
- 15,000 variants (10,500 / 2,100 / 2,400)

Bins

height ≈ 75 / 96 / 127 px
Tilt ≈ 16 / 90 / 92

Pipeline


Image
YOLO11s
Top crop
Preproc. (optional)
CRNN
CTC decode
Conf. gate τ=0.50
Text / abstain

Pipeline

- Detection is single-class YOLO11s at 640 px. Only the top-score box is kept
- Crop is axis-aligned rectangle, no padding, no tilt correction
- Recognition: CRNN on image scaled to 48×256 canvas with aspect-preserving fit and padding; CTC over 58 characters
- Confidence per character is max probability in CTC decoding run; τ=0.50

Detection: fine-tune YOLO11s on the task

- Single class: license plate
- 640 px input, batch 16, AdamW, lr 0.002
- 3,500 original-scale images, 24 epochs
- Best checkpoint by validation mAP@0.5 (epoch 4)

Preprocessing

- Three modes: `none` / grayscale + CLAHE (clip 2.0, 8×8) / Canny
- Applied at inference time, after the crop and before the recognizer
- The CRNN trains on raw crops, so any other mode is a distribution shift away from training

Recognition: CRNN replaces EasyOCR

- CNN + bidirectional LSTM + CTC, 2.99M parameters, selected epoch 39
- Alphabet: 58 characters + CTC blank
- 3,500 crops, total for training 50 epochs
- No preprocessing by default, confidence threshold 0.50

CTC decoding

- The CRNN emits, at each of 16 timesteps, a probability over 59 symbols (58 characters + blank)
- Greedy CTC decode: collapse repeated symbols, then remove the blanks
- Per-character confidence is the maximum probability inside its decoding run
- These per-character confidences feed the plate-level mean used by the gate

Confidence

- Each decoded character carries a probability (max over its CTC run)
- Plate score is mean of the character probabilities
- Threshold τ chosen on validation only: sweep 0.50–0.99, maximize (accepted exact accuracy + coverage -1)
- At the selected $\tau = 0.50$ all 800 test plates pass (lowest mean confidence 0.52)
- Raising τ to 0.90 abstains 259 plates. This raises number of false negatives in P/R/F1

Fine-tuning the detector closes the end-to-end gap

Config	mAP@0.5	F1	Char. acc.	Exact
Frozen YOLO11s + EasyOCR	0.9073	0.9167	0.3240	0 / 800
CRNN, GT crops	–	–	0.7593	267 / 800
CRNN + frozen YOLO11s	0.9073	0.9167	0.6710	148 / 800
CRNN + fine-tuned YOLO11s	0.9949	0.9714	0.7661	296 / 800

Canny preprocessing destroys the signal

Preprocessing (val. char. acc.)	CRNN
None	0.7558
Grayscale + CLAHE	0.7469
Canny edge detection	0.4575

Original (GT: 浙JD389L)



Canny (prediction: 皖AU0367)



Accuracy holds across scales

Scale	Det. mAP@0.5	Char. acc.	Exact plates
1.00	0.9949	0.7661	296 / 800
0.75	0.9966	0.7689	300 / 800
0.50	0.9910	0.7174	241 / 800

- OCR accuracy doesn't depend on absolute size

Failure analysis

Model	Prediction (ground truth)	Likely reason
EasyOCR	苏 E050R8 → 05oRo 菲	general OCR, no plate prior
CRNN	皖 AR1654 → 皖 AR1564	large crop shrunk to 48×256
Frozen YOLO11s	51 of 800 missed	not trained on CCPD scenes

(a) 苏E050R8 read as 05oRo菲 (EasyOCR)



(b) 皖AR1654 read as 皖AR1564 (CRNN, large plate)



(c) plate missed by frozen YOLO (red box = ground truth)



Missed detections drive end-to-end error

- Frozen pipeline: 0.6710 vs. 0.7593 on GT crops
- Fine-tuning recovers the gap: 0.7661
- EasyOCR: 0 / 800 plates exact
- Fixed 48×256 canvas: OCR is scale-invariant
- Next: larger inputs, size/tilt-balanced training

Contributors

- Andrei Polevoi — data and preprocessing
- Maximilian Mifsud Bonici — detection
- Evgenii Khovalyg — recognition
- Mikhail Krylov — pipeline and evaluation

Questions?

Thank you

https://github.com/S1norin/inno-detecting-license-plates

Demo video: https://drive.google.com/file/d/1KBiiD-dvhvzsI7yWwDc7KqwM6jp585ez/view